Sean Luka Girgis

Senior Data Engineer | Lakehouse & AI Data Platform

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience and deep strength in Python, SQL, Spark/PySpark, data modeling, ETL pipelines, reconciliation, and production data reliability. Built financial-services data platforms and telemetry pipelines supporting large-scale analytics, forecasting, reporting, and operational decision-making. Strong fit for lakehouse and AI data platform work that requires reliable ingestion, curated datasets, schema evolution discipline, partitioning strategy, data quality controls, documentation, and pragmatic engineering tradeoffs. Currently extending AWS/PySpark experience into modern lakehouse tools including Apache Iceberg, Delta Lake, Databricks, Snowflake, and dbt through hands-on study and project work.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Architected and supported Python/Pandas and PySpark-oriented ETL pipelines ingesting P95 telemetry from 6,000+ infrastructure endpoints, turning operational source data into curated datasets for reporting, analytics, and capacity forecasting.
- Designed data models, Oracle schemas, historical retention structures, and reporting-ready layers that supported trend analysis, seasonal risk forecasting, and executive technology decision-making in a regulated banking environment.
- Built a hybrid AWS/Oracle data platform using S3 landing patterns, AWS Glue ETL, Athena/Redshift concepts, and production-grade validation practices to improve scalability and support cloud-based forecasting workloads.
- Implemented data quality, reconciliation, and root-cause analysis practices across telemetry feeds from BMC TrueSight, CA Wily, AppDynamics, and related monitoring sources to improve trust in downstream analytics outputs.
- Optimized pipeline and query performance through partitioning concepts, schema design, file format awareness, and cost-conscious processing patterns across large operational datasets.
- Documented pipeline logic, data flows, support procedures, and operational runbooks; partnered with engineering and stakeholder teams to resolve production data issues and improve repeatability.
- Developed ML forecasting workflows using Prophet and scikit-learn to predict capacity bottlenecks six months ahead, connecting reliable data engineering foundations to analytics and AI-style decision support.
- Mentored and guided team members on data interpretation, operational reporting, pipeline troubleshooting, and disciplined documentation in a high-availability financial-services environment.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics workflows around Dynatrace AppMon, Gomez Synthetic Monitoring, and user-performance data to identify performance patterns and produce optimization recommendations for critical digital systems.

- Led the FAST project, mining real-user and synthetic monitoring datasets to surface actionable recommendations adopted by engineering teams for performance and reliability improvement.
- Supported AWS migration planning and APM readiness by analyzing monitoring data, transaction traces, infrastructure behavior, and operational risks across web and mobile workloads.
- Created stakeholder-facing reports and dashboards that translated raw telemetry into business and engineering insights for non-technical and technical audiences.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM SME for a large financial-services client, supporting enterprise-scale monitoring environments with 50+ Enterprise Managers and thousands of instrumented agents.
- Developed Perl/Korn shell and SQL-driven reporting automation that extracted, transformed, and delivered operational performance data for recurring analysis and support workflows.
- Created dashboards, alerts, SLA thresholds, technical documentation, and standards that improved operational support and consistency across distributed teams.
- Collaborated with J2EE/WebLogic and database teams to investigate performance issues, analyze telemetry, and support root-cause resolution across enterprise applications.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, supporting transaction volumes described as exceeding VISA-scale throughput.
- Automated schema conversion, validation, and migration verification processes to ensure data correctness and application continuity during phased cutover.
- Optimized core C++/OCCI transaction processing and database access paths, reducing hardware footprint while maintaining sub-second query latency.
- Built testing frameworks and reconciliation checks to validate migrated records and preserve confidence in production data outputs.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed and supported high-availability telecom billing interfaces using C/C++, Pro*C, PL/SQL, SQL scripts, XML, sockets, and UNIX shell automation across HPUX, SunOS, and Linux environments.
- Improved performance and reliability of billing and reporting processes through database tuning, sequence optimization, memory reduction, and throughput-focused engineering changes.
- Built and maintained production interfaces, reverse-engineered undocumented APIs, and documented integration behavior to support long-running enterprise billing workflows.
- Led QA cycles and production bug triage using structured test management practices, reinforcing strong habits around validation, defect tracking, and release quality.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, Java, C++, C, PL/SQL, Perl, Korn Shell, XML, JSON

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed and documented a serverless AWS data platform pattern using S3, Glue, Athena, Redshift, Parquet, and Bedrock concepts to support analytics, AI data access, and cost-aware query design.
Project work emphasizes raw-to-curated data flow, partitioning, schema evolution awareness, metadata, and reliable data access patterns.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, Bedrock

HorizonScale — AI Capacity Forecasting Engine
Built a Python/PySpark forecasting and analytics workflow for capacity planning, transforming operational telemetry into forecast-ready datasets and stakeholder dashboards.
The work connects data engineering, quality checks, historical modeling, and AI/ML-ready data preparation.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Pipelines

AI-Powered Job Search Pipeline
Built an agentic Python automation pipeline with semantic duplicate detection, structured JSON artifacts, quality gates, and tracking.
Uses RAG-style retrieval patterns, embeddings, FAISS, LLM APIs, and reproducible workflow design to support data-driven application decisions.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG Concepts, JSON Contracts, Automation